

Notes: Asker, Collard-Wexler, and De Loecker (AER, 2019)

(Mis)Allocation, Market Power, and Global Oil Extraction

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1 Big picture

1.1 Main question

The paper asks how much production is misallocated in the global crude oil extraction industry and how much of that misallocation can be attributed to market power. The central idea is simple: if low-cost oil fields delay production while high-cost fields produce, the world uses too many resources to obtain the same quantity of oil. The paper turns this idea into a welfare-based measurement framework.

1.2 Core answer

The authors find substantial productive inefficiency in global oil extraction. In their baseline dynamic calculation, the net present value of actual extraction costs is about \$2.5 trillion in 2014 dollars, while the competitive counterfactual cost is about \$1.76 trillion. The difference, about \$744 billion, is the total measured productive inefficiency. A lower-to-upper bound of roughly \$105 to \$163 billion is attributed to OPEC-related market power, corresponding to about 14.1% to 21.9% of total measured misallocation.

1.3 Why this paper matters

Most misallocation papers use dispersion in revenue productivity, or TFPR, to infer distortions. This paper instead uses direct cost data and compares realized cost curves with counterfactual efficient supply curves. This matters because:

- the measure maps directly into welfare-relevant resource costs;
- the approach separates productive inefficiency from the standard deadweight-loss triangle;
- the method can decompose total misallocation into specific sources, such as market power, taxes, wars, regulation, and other frictions.

2 Incremental contribution of the paper

2.1 Relative to TFPR-based misallocation measurement

The paper's first contribution is methodological. Rather than measuring misallocation through dispersion in revenue productivity, the authors build a cost-based measure of productive inefficiency. The key object is the difference between actual extraction costs and the costs that would arise if the same aggregate production path were produced by price-taking firms.

2.2 Relative to direct and indirect approaches

The paper combines the two main approaches in the misallocation literature.

- The **indirect approach** estimates total misallocation by comparing actual outcomes with a model-implied efficient benchmark.
- The **direct approach** studies specific distortions, such as taxes, market power, or financial frictions.

This paper first estimates total misallocation and then decomposes it into components attributable to OPEC-related market power and other sources.

2.3 Relative to industrial organization and natural-resource economics

The paper contributes to industrial organization by measuring the welfare cost of market power through a production allocation channel. It contributes to natural-resource economics by embedding finite resource extraction into a misallocation framework. The model must account for the fact that oil extraction is dynamic: extracting a low-cost field today changes what is available tomorrow.

2.4 The main conceptual novelty

The most important conceptual move is to measure **productive inefficiency** rather than output restriction. Market power can create two welfare losses:

1. a deadweight-loss triangle because output is restricted;
2. a productive-inefficiency rectangle because production is shifted from low-cost to high-cost units.

The paper focuses on the second object, which is directly about misallocation across producers.

3 Data and setting

3.1 Unit of analysis

The unit of analysis is the oil field. The paper uses detailed Rystad Energy data on 13,248 active oil fields from 1970 to 2014. The data cover roughly 92% of world production and include annual production, reserves, operating costs, capital costs, field characteristics, location, geology, and technology type.

3.2 Why oil is a useful setting

The oil market is unusually useful for studying cost-based misallocation.

- Oil is close to a homogeneous commodity, so cross-field cost differences are not mainly output-quality differences.
- Production costs vary greatly for geological reasons. A Saudi onshore field and a Norwegian offshore field can have very different resource costs.
- Low-cost OPEC fields have large reserves and often extract more slowly than higher-cost non-OPEC fields.
- The existence of OPEC provides an observable institutional source of market power.

3.3 Cost measure

The baseline cost measure sums expenditure categories related to extraction:

- well capital expenditures;
- facility capital expenditures;

- abandonment costs;
- production operating expenditures;
- transportation operating expenditures;
- selling, general, and administrative expenses.

Taxes, royalties, and government profit oil are excluded in the baseline resource-cost measure, but the paper studies robustness to including them.

3.4 Reserves and production

The paper uses oil reserves to account for the finite-resource constraint. In the baseline, reserves are based on P50 estimates at an oil price of \$70 per barrel, plus actual production history. The counterfactual allocation cannot simply move all low-cost oil to the present if reserves or extraction-rate constraints prevent it.

4 Measurement framework

4.1 Productive inefficiency

The paper defines productive inefficiency as the net present value of the difference between actual production costs and the costs of producing the realized quantity path under price-taking behavior. Formally, the key comparison is:

$$\text{Productive inefficiency} = \text{NPV}(\text{actual resource costs}) - \text{NPV}(\text{counterfactual efficient resource costs}).$$

The aggregate quantity of oil produced each year is held fixed. The paper asks whether that quantity could have been produced by a lower-cost allocation of fields.

4.2 Why this differs from standard market-power welfare loss

In a static monopoly diagram, market power reduces quantity and creates a deadweight-loss triangle. But if low-cost producers restrict output and high-cost producers fill the gap, there is also a productive inefficiency: the same quantity is produced too expensively. The paper focuses on this latter object.

4.3 Production technology

The paper models field-level production with a Leontief technology:

$$q_{ft} = \min\{\alpha_{ft}K_{ft}, \gamma_{ft}L_{ft}\},$$

subject to the reserve constraint

$$q_{ft} \leq R_{ft}, \quad R_{ft} = R_{ft-1} - q_{ft-1}.$$

Under cost minimization, the field has constant marginal cost up to the reserve or extraction constraint. This makes average cost informative about marginal cost.

4.4 Competitive benchmark

In the competitive benchmark, all fields take prices as given. The key result is that lower-cost fields should be extracted before higher-cost fields, subject to reserve and extraction constraints. Thus, when high-cost fields produce while low-cost fields with remaining reserves are held back, the pattern reveals productive inefficiency.

4.5 Sorting algorithm

The counterfactual allocation is constructed by a sorting algorithm:

1. In each year, sort available oil fields from lowest to highest marginal cost.
2. Produce from the lowest-cost field until either its reserve or extraction limit binds or aggregate demand for that year is met.
3. Move to the next-lowest-cost field and repeat.
4. Update reserves and continue to the next year.

This produces the dynamic cost-minimizing path for the observed aggregate oil quantity path.

5 Market power decomposition

5.1 Why OPEC matters

OPEC is treated as the main observable source of market power. OPEC members include large low-cost producers such as Saudi Arabia and Kuwait. If OPEC restricts production, production shifts to higher-cost non-OPEC fields, creating productive inefficiency.

5.2 Lower and upper bounds

The paper does not attribute all misallocation to market power. Instead, it constructs bounds:

- The **lower bound** attributes only the distortion between OPEC and non-OPEC production to OPEC market power.
- The **upper bound** also includes cross-country distortions within OPEC, since inefficient cartel quota allocations may be part of how OPEC sustains market power.

This is conservative because many non-OPEC production decisions may also respond strategically to OPEC behavior.

5.3 Other sources of misallocation

The remaining productive inefficiency may come from:

- taxes and royalties;
- local regulation and environmental restrictions;
- national security or energy-independence objectives;
- wars, sanctions, and infrastructure destruction;

- political economy and corruption;
- measurement error and expectation errors about reserves and field costs.

The paper explicitly discusses these sources rather than treating the residual as a single unexplained friction.

6 Main empirical results

6.1 Magnitude of total misallocation

The baseline dynamic calculation gives a large welfare-relevant resource-cost distortion. From 1970 to exhaustion, actual costs are about \$2.499 trillion, while the counterfactual efficient path costs about \$1.756 trillion. The implied total distortion is about \$744 billion.

6.2 Contribution of OPEC market power

The paper attributes between \$105 billion and \$163 billion of the distortion to OPEC-related market power. This equals roughly 14.1% to 21.9% of total measured productive inefficiency. Thus, market power is quantitatively important, but it is not the only distortion in global oil extraction.

6.3 Static evidence

The static decomposition shows that distortions are especially large when oil prices are high. When prices rise, high-cost producers enter or expand, so the resource-cost loss from withheld low-cost production becomes larger. After 2008, OPEC-related components account for a large share of annual static distortions.

6.4 Country-level patterns

Saudi Arabia and Kuwait have low-cost production and large reserves. In a competitive counterfactual, they would produce much more early in the sample. By contrast, high-cost fields in the United States, Canada, Norway, Venezuela, and Nigeria often produce despite the availability of lower-cost OPEC reserves. This is the central empirical pattern behind the estimated misallocation.

6.5 Robustness

The results are robust to:

- changing extraction-rate limits;
- using P90 instead of P50 reserves;
- including distortionary taxes and royalties;
- restricting the sample to fields already active in 1970;
- changing the assumed first year of field availability;
- considering inferred wedges from unobserved distortions;
- examining start-up costs, marginal-cost curvature, and alternative discount-factor interpretations.

7 Interpretation

7.1 Economic mechanism

The economic mechanism is an intertemporal reallocation problem. Low-cost OPEC producers restrict or delay production. Higher-cost fields elsewhere then satisfy current demand. Since future costs are discounted, delaying low-cost extraction and bringing forward high-cost extraction raises the present value of resource costs.

7.2 Why cost data are powerful

Direct cost data allow the paper to measure welfare losses more transparently than TFPR-based measures. TFPR dispersion can reflect markups, demand differences, measurement error, or quality variation. In this setting, oil is relatively homogeneous and field-level extraction costs are directly observed, making the counterfactual clearer.

7.3 Relation to the theory of second best

The paper recognizes that distortions interact. Removing OPEC market power while leaving taxes, wars, and other distortions in place may not mechanically equal the difference between the fully efficient allocation and actual allocation. The paper therefore distinguishes infra-marginal measurement from a marginal second-best exercise using inferred wedges.

8 Strengths of the paper

8.1 Clear welfare object

The paper’s main strength is that its misallocation measure has a clear welfare interpretation: it is the excess resource cost of producing the observed oil quantities. This makes the result economically meaningful and directly comparable to other welfare-cost estimates.

8.2 Rich global micro-data

The data cover a large share of global oil production at the field level. This makes it possible to study production allocation across thousands of productive units rather than relying only on country-level aggregates.

8.3 Careful decomposition

The paper does not simply claim that OPEC causes all inefficiency. It decomposes total distortion into within-country, across-country, within-OPEC, and between-OPEC/non-OPEC components. This makes the market-power attribution much more credible.

8.4 General methodological value

The paper offers a template for using cost curves to measure misallocation in other industries. The method is especially useful where producer-level cost data exist and where aggregate output can be held fixed while production is reallocated across units.

9 Limitations and caveats

9.1 Measurement error

Field-level costs and reserves are estimated from heterogeneous sources. If low-cost and high-cost fields are mismeasured, the algorithm may infer misallocation where none exists. The paper conducts robustness checks, but measurement error cannot be fully ruled out.

9.2 Expectation error

Operators do not know future reserves and costs perfectly. Some production patterns that look inefficient ex post may have been reasonable ex ante. This is especially relevant for fields whose costs or reserves are highly uncertain at discovery.

9.3 Simplified conduct model

The paper does not fully model OPEC as a dynamic cartel with bargaining, quotas, cheating, geopolitics, and strategic reactions by non-OPEC producers. Instead, it uses OPEC as an observed institutional source of market power and constructs bounds.

9.4 Constant marginal cost approximation

The baseline assumes constant marginal costs up to extraction constraints. The paper checks curvature and argues that it does not drive the results, but this remains a simplifying assumption.

10 How to use this paper in future research

10.1 For misallocation research

This paper is useful as an alternative to TFPR-based misallocation. If a researcher has detailed cost data, the paper suggests measuring misallocation through excess production costs rather than productivity residual dispersion.

10.2 For market-power research

The paper shows that market power may matter not only through price-quantity deadweight loss but also through productive inefficiency. A cartel can distort which firms or fields produce, not just how much is produced.

10.3 For natural-resource and energy economics

The model gives a clean way to think about the order of extraction. In an undistorted world, cheaper reserves should be extracted before expensive reserves, subject to physical constraints. Deviations from this order are informative about distortions.

11 Extracted figures and tables from the paper

11.1 Figure 1: Production Misallocation (Resulting from Market Power) and Figure 2: OPEC Market Share and Oil Price, 1970–2014

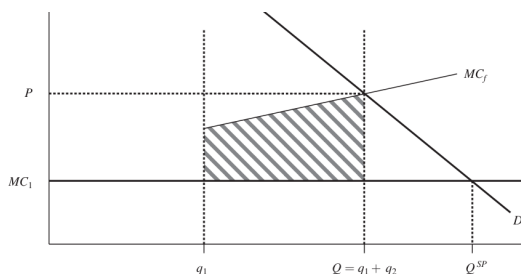


FIGURE 1. PRODUCTION MISALLOCATION (RESULTING FROM MARKET POWER)

q_1 indicates total production from the cartel, while q_f indicates production from the competitive fringe with market power has marginal costs of MC_c , while the fringe has the marginal cost schedule of the social planner's quantity.

Figure 1: Production Misallocation (Resulting from Market Power).

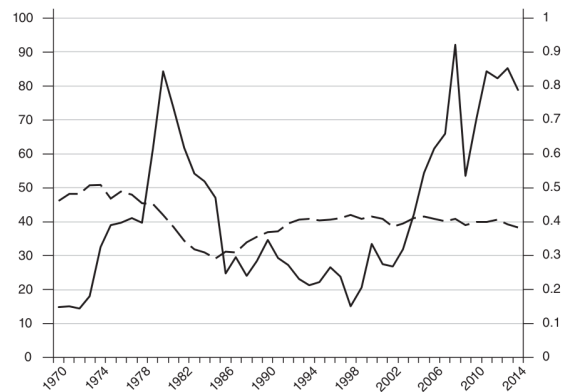


FIGURE 2. OPEC MARKET SHARE AND OIL PRICE, 1970–2014

The vertical axis on the left is in dollars and corresponds to the annual average oil price, which is in black line. This price series is deflated with the US GDP deflator (base year 2009). The OPEC market share is indicated by the dashed black line. The vertical axis on the right indicates the level of the countries included in OPEC in all years if they had ever had active membership between 1970 and 2014.

Figure 2: OPEC Market Share and Oil Price, 1970–2014.

Read. Figure 1 separates the productive-inefficiency rectangle from the standard deadweight-loss triangle. Figure 2 shows the evolution of OPEC market share and oil prices over 1970–2014.

Economic message. Market power matters not only because it can reduce total output, but also because it reallocates production from low-cost to high-cost suppliers. The OPEC share series motivates the market-power channel used in the decomposition.

11.2 Table 1: Largest Crude Producers, Percent of Global Production, 1970–2014 and Table 2: Summary Statistics, by Field-Year

TABLE 1—LARGEST CRUDE PRODUCERS, PERCENT OF GLOBAL PRODUCTION, 1970–2014

OPEC		Non-OPEC	
Saudi Arabia	11.8	United States	14.4
Iran	5.4	Russia	13.0
Venezuela	3.8	China	4.1
UAE	3.1	Mexico	3.7
Nigeria	2.8	Canada	3.3
Iraq	2.7	United Kingdom	2.4
Kuwait	2.6	Norway	2.4

Notes: Global production from 1970–2014 was 1,156 billion barrels. Collectively, these 14 countries account for 85.4 percent of global production.

Table 1: Largest Crude Producers, Percent of Global Production, 1970–2014.

TABLE 2—SUMMARY STATISTICS, BY FIELD-YEAR

	Mean	Median	5%	95%
<i>Field-year characteristics</i>				
Production (mB/year)	3.43	0.22	0.00	10.92
Reserves (mB)	99.49	3.71	0.03	239.78
Discovery year	1965	1967	1911	1999
Startup year	1971	1974	1916	2005
Off-shore	0.19			
<i>Costs (\$m)</i>				
Exploration capital expenditures	0.61	0.00	0.00	0.41
Well capital expenditures	9.10	0.49	0.00	35.32
Facility capital expenditures	5.14	0.21	0.00	16.85
Production operating expenditures	10.41	0.46	0.00	38.47
Transportation operating expenditures	2.27	0.13	0.00	7.01
SGA operating expenditures	2.65	0.22	0.00	8.85
Taxes operating expenditures	1.41	0.00	0.00	1.09
Royalties	18.19	0.40	0.00	45.36
Government profit oil	15.59	0.00	0.00	21.00

Notes: Only fields with active production during 1970–2014 are included. There are 66,920 fields in the Rystad data. 13,248 of these fields have active production. Reserves data exists for 13,298 fields. As a result, in Section V, 11,457 fields are used. All numbers are in \$US deflated by the US GDP deflator for 2009. mB indicates million barrels. The unit of observation is the field-year.

Table 2: Summary Statistics, by Field-Year.

Read. Table 1 shows that Saudi Arabia, Russia, and the United States are central producers, but their institutional structures differ sharply. Table 2 documents the field-year data, including production, reserves, discovery years, offshore status, and cost categories.

Economic message. The empirical setting has both concentrated geopolitical actors and rich micro-level field heterogeneity. This combination makes it possible to connect market power to resource-cost misallocation.

11.3 Figure 3: Production Costs and Price, 1970–2014: OPEC Countries and Figure 4: Production Costs and Price, 1970–2014: Other Countries

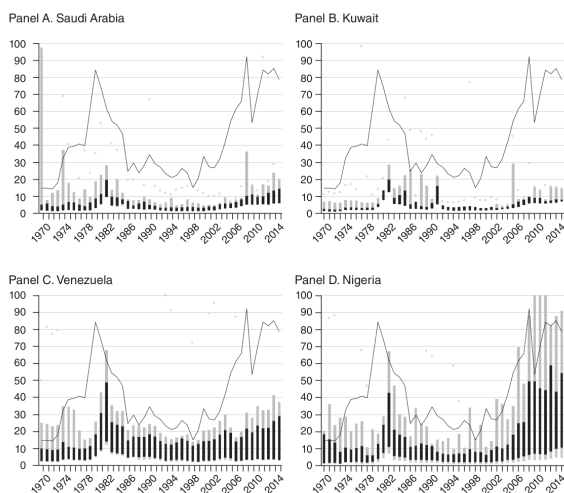


FIGURE 3. PRODUCTION COSTS AND PRICE, 1970–2014: OPEC COUNTRIES

Notes: Each panel plots the dispersion of the costs of production (by barrel) in a country, and the price of oil. The vertical axis is \$/barrel, from 0 to 100 in increments of 10. The horizontal axis is in years, from 1970 to 2014. Costs are indicated by the bars and circles. The (gray and black) bar indicates the range of costs within the first and ninety-ninth percentiles of production. That is, the cheapest, and most expensive, 1 percent of barrels produced in the year are excluded. The black portion of the bar indicates the fifth to ninety-fifth percentiles range. Circles indicate the maximum cost per barrel incurred in a year. Where a cost exceeds \$100 per barrel, it is not shown (the vertical axis is truncated at 100). This accounts for many of the maxima not being visible, for instance. The oil price is indicated by the black line. All series have been deflated with the US GDP deflator (base year 2009). All costs are measured according to the baseline specification.

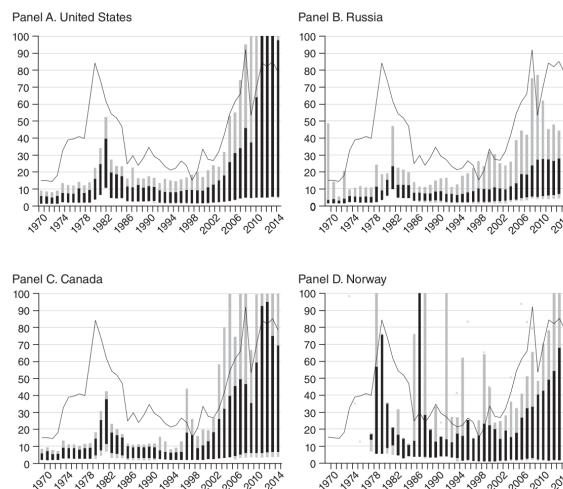


FIGURE 4. PRODUCTION COSTS AND PRICE, 1970–2014: OTHER COUNTRIES

Notes: The notes for Figure 3 also apply to this figure. The Norwegian panel reflects little meaningful production prior to 1978.

Figure 4: Production Costs and Price, 1970–2014: Other Countries.

Figure 3: Production Costs and Price, 1970–2014: OPEC Countries.

Read. Figure 3 shows low and relatively tight costs in Saudi Arabia and Kuwait, but higher and more dispersed costs in Venezuela and Nigeria. Figure 4 shows broader cost dispersion among non-OPEC producers, especially the United States, Canada, Russia, and Norway.

Economic message. The figures provide the central descriptive evidence: low-cost OPEC fields coexist with high-cost production elsewhere. This creates the potential for large welfare gains from reallocation.

11.4 Table 3: Reserves and Production, 2014 and Table 4: Unit Costs across the Global Oil Industry, 1970–2014

TABLE 3—RESERVES AND PRODUCTION, 2014

	Reserves (mB)	Share of world reserves (percent)	Reserves Annual production (percent)
Non-OPEC	218,054	50	10
Russia	46,134	11	12
Canada	36,622	8	43
United States	31,735	7	7
Norway	6,962	2	10
OPEC	220,561	50	19
Saudi Arabia	74,194	17	18
Venezuela	17,523	4	19
Kuwait	15,723	4	16
Nigeria	7,952	2	10

Notes: Data are for 2014. Total reserves for the world in 2014 were 438 billion barrels. The ratio of reserves-to-production was 14. OPEC countries are listed in Section IIB. Countries are included in OPEC in all years if they had ever had active membership between 1970 and 2014. Reserves are reported using P50 measures at a world price of \$70 per barrel.

Table 3: Reserves and Production, 2014.

TABLE 4—UNIT COSTS ACROSS THE GLOBAL OIL INDUSTRY, 1970–2014

	1970–1979	1980–1989	1990–1999	2000–2014
Number of active fields	4,766	7,088	9,760	12,085
Mean oil price	20	40	21	59
Mean global production (mB/year)	20,861	21,489	23,984	26,298
OPEC	9,979	7,289	9,606	11,249
Mean global reserves (mB)	737,928	728,532	661,815	517,559
OPEC	392,912	365,891	328,914	254,730
Unit costs (baseline specification):				
95th percentile Saudi Arabia	5.8	13.6	4.4	10.4
Median Saudi Arabia	2.3	5.6	2.3	5.4
95th percentile OPEC	6.7	18.6	7.6	20.1
Median OPEC	2.4	5.9	2.8	6.1
95th percentile non-OPEC	6.7	15.6	9.2	28.2
Median non-OPEC	3.6	7.0	4.1	9.7
Unit costs (including taxes and royalty payments):				
95th percentile Saudi Arabia	5.8	13.6	4.4	10.4
Median Saudi Arabia	2.3	5.6	2.3	5.4
95th percentile OPEC	30.2	53.6	21.1	79.1
Median OPEC	2.8	13.6	6.5	12.0
95th percentile non-OPEC	26.3	40.1	20.3	75.3
Median non-OPEC	9.1	14.8	9.1	24.0

Notes: The unit cost is computed as per Section IVA and top-coded at \$100. The unit of observation for unit cost is the barrel. Percentiles and medians are calculated at the barrel level. All prices and costs are deflated with the US GDP deflator (base year 2009). Reserves are reported using P50 measures at a world price of \$70 per barrel. OPEC countries are listed in Section IIB. Countries are included OPEC in all years if they had ever had active membership between 1970 and 2014. Only fields active between 1970 and 2014 are included.

Table 4: Unit Costs across the Global Oil Industry, 1970–2014.

Read. Table 3 shows that OPEC has about half of world reserves in 2014 and a higher reserves-to-production ratio than non-OPEC. Table 4 compares costs over time for Saudi Arabia, OPEC, and non-OPEC production.

Economic message. OPEC does not lack reserves. The fact that low-cost OPEC producers hold large reserves while high-cost fields elsewhere produce is consistent with strategic production restraint and other distortions.

11.5 Figure 5: OPEC Market Share, 1970–2014 and Table 5: Dynamic Counterfactual Results

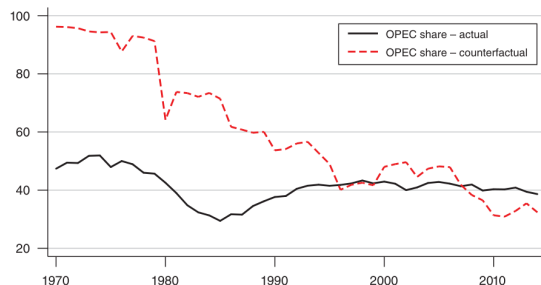


FIGURE 5. OPEC MARKET SHARE, 1970–2014

OPEC share – counterfactual presents the share of production accounted for by OPEC.

Figure 5: OPEC Market Share, 1970–2014.

TABLE 5—DYNAMIC COUNTERFACTUAL RESULTS (NPV OF COSTS IN BILLIONS OF 2014 US\$)

	Timespan	
	1970–2014	1970–2100
Actual (A)	2,184 (125)	2,499 (130)
Counterfactual (C)	1,268 (76)	1,756 (79)
Total distortion (A – C)	916 (124)	744 (112)
Decomposition of total distortion		
Within country (non-OPEC)	329 (80)	284 (41)
Within country (OPEC)	192 (46)	157 (72)
Across country (within non-OPEC)	163 (18)	139 (17)
Across country (within OPEC) (X)	85 (22)	58 (21)
Between OPEC and non-OPEC (Y)	148 (29)	105 (25)
Production distortion due to OPEC market power		
Upper bound (X + Y)	233 (42)	163 (38)
Lower bound (Y only)	148 (29)	105 (25)

Notes: The NPV of costs from 1970 to 2014, and to 2100 (exhaustion of all fields), are reported in billions of 2014 dollars (assuming a 5 percent discount rate). Results are for the baseline specification: a field extraction rate of 10 percent of reserves is imposed in the counterfactual; the P50 measures of reserves are used where needed, and a demand growth rate of 1.3 percent per year after 2014 is assumed. The Actual path is that observed in the data. The Counterfactual path is that computed using the unconstrained sorting algorithm. The within-country (non-OPEC) decomposition takes the path from the sorting algorithm in which all non-OPEC countries are constrained to produce their actual production. OPEC fields produce as in the data. The reported number is $A -$ [the NPV of the costs of this path] = D1. The within-country (OPEC) decomposition is the mirror of this for OPEC countries (= D2). The across-country (within non-OPEC) decomposition takes the path from the sorting algorithm in which non-OPEC production is constrained to match the observed amount. OPEC fields produce as in the data. The reported number is $A - D1 -$ [the NPV of the costs of this path] = E1. The across-country (within OPEC) decomposition is the mirror of this for OPEC countries (= E2). The Between OPEC and non-OPEC decomposition takes the path from the unconstrained sorting algorithm. The reported number is $A - D1 - D2 - E1 - E2 - C = F1$. Bootstrapped standard errors in parentheses using 50 bootstrap replications.

Table 5: Dynamic Counterfactual Results (NPV of Costs in Billions of 2014 US dollars).

Read. Figure 5 compares actual OPEC share with the competitive counterfactual share. The counterfactual assigns much more early production to OPEC. Table 5 reports the baseline NPV calculations and decomposition.

Economic message. The dynamic counterfactual shows that the efficient allocation would have used low-cost OPEC reserves much earlier. The cost gap between actual and counterfactual paths is the main quantitative result.

11.6 Figure 6: Decomposing Static Distortions and Figure 7: Within-Country Deviations From Cost-Minimizing Production

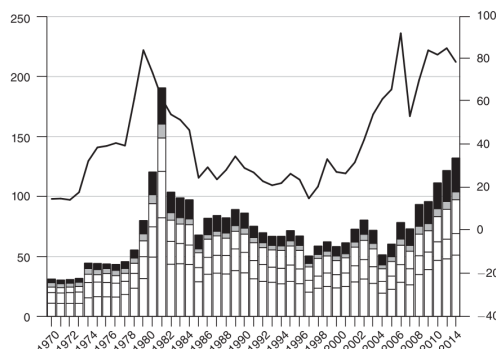


FIGURE 6. DECOMPOSING STATIC DISTORTIONS

Notes: Static distortions for each year are presented in 2014 dollars (left vertical axis), with the total height of each bar representing the difference between the actual cost of production and the optimal cost of production (the total distortion). Each bar is decomposed into the following distortions (from bottom to top): Within-country (OPEC); Within-country (non-OPEC); Across country (Within non-OPEC); Across-country (within OPEC) (grey); Between-OPEC, and non-OPEC (in black). Definitions of distortions decompositions mirror those in Table 5, although only applying to the individual year of interest. The oil price is shown using the black line dollar corresponding to the right vertical axis.

Figure 6: Decomposing Static Distortions.

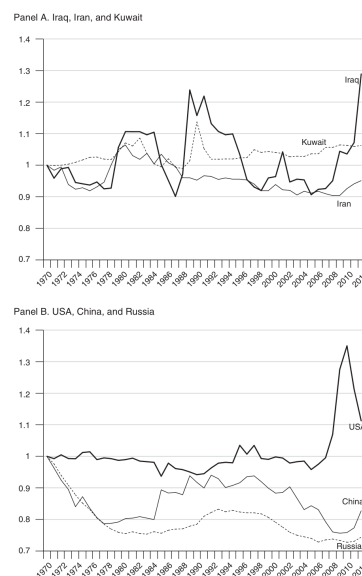


FIGURE 7. WITHIN-COUNTRY DEVIATIONS FROM COST-MINIMIZING PRODUCTION: SELECTED COUNTRIES, 1970-2014

Notes: In panel A, Iraq is heavy bold, Iran is light bold line, and Kuwait is the dashed line. In panel B, USA is heavy bold, China is light bold line, and Russia is the dashed line. The ratio of total observed cost and cost-minimizing (optimal) cost, taking the actual production up to that year as an initial condition (as per static distortion calculations) and holding country production fixed, is computed for each year (i.e., annual within-country deviations from cost minimization). This ratio is indexed to the level in 1970. The indexed ratio (vertical axis) is graphed by year (horizontal axis). The ratio in 1970 (i.e., the base year) for Iraq, Iran, Kuwait, United States, China, and Russia is 1.385, 1.563, 1.037, 1.401, 1.956, and 1.742 respectively.

Figure 7: Within-Country Deviations From Cost-Minimizing Production: Selected Countries, 1970-2014.

Read. Figure 6 decomposes annual static distortions into within-country, across-country, within-OPEC, and between-OPEC/non-OPEC components. Figure 7 tracks within-country deviations for selected countries.

Economic message. Distortions are not constant over time. They rise when oil prices are high and around geopolitical disruptions. Wars and institutional transitions matter alongside cartel behavior.

11.7 Table 6: Static Counterfactual for 2014: Top 20 Producers and Table 7: Dynamic Counterfactual Results, Alternate Specifications

TABLE 6—STATIC COUNTERFACTUAL FOR 2014: TOP 20 PRODUCERS

Country	Actual output share	Counterfactual output share	Δ Share
Persian Gulf OPEC	0.258	0.744	0.486
Iran	0.057	0.091	0.034
Iraq	0.029	0.069	0.040
Kuwait	0.030	0.155	0.125
Qatar	0.009	0.015	0.006
Saudi Arabia	0.133	0.414	0.281
United Arab Emirates	0.031	0.075	0.044
Other OPEC	0.135	0.044	-0.091
Algeria	0.021	0.015	-0.006
Indonesia	0.020	0.002	-0.018
Libya	0.025	0.012	-0.013
Nigeria	0.028	0.006	-0.022
Venezuela	0.041	0.009	-0.032
Non-OPEC	0.607	0.212	-0.395
Brazil	0.014	0.001	-0.013
Canada	0.023	0.006	-0.017
China	0.045	0.002	-0.043
Kazakhstan	0.010	0.000	-0.01
Mexico	0.023	0.013	-0.01
Norway	0.027	0.009	-0.018
Russia	0.144	0.047	-0.097
United Kingdom	0.022	0.001	-0.021
United States	0.132	0.013	-0.119
Rest of the world	0.136	0.044	-0.092

Notes: Reported results are for the top 20 producers between 1970 and 2014. Initial conditions are the state of the global market at the end of 2013. Application of the sorting algorithm gives counterfactual production for 2014. In every other respect, the baseline specification is used: a field extraction rate of 10 percent of reserves is imposed in the counterfactual; the P50 measures of reserves are used where needed; and a demand growth rate of 1.3 percent per year after 2014 is assumed.

TABLE 7—DYNAMIC COUNTERFACTUAL RESULTS, ALTERNATE SPECIFICATIONS

	Specification							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Actual (A)	2,499	2,467	2,507	2,499	4,484	2,474	1,465	2,500
Counterfactual (C)	1,756	1,804	1,713	1,757	2,839	1,703	1,021	1,797
Total distortion (A - C)	744	664	793	742	1,645	771	444	703
Proportion: (A - C)/A	0.298	0.269	0.316	0.297	0.367	0.312	0.303	0.281
Distortion due to OPEC								
Upper bound (X + Y)	163	148	150	161	747	196	179	188
Lower bound (Y only)	105	89	95	104	225	99	120	125
Proportion: (X + Y)/(A - C)	0.219	0.224	0.189	0.218	0.454	0.255	0.404	0.268
Proportion: Y/(A - C)	0.142	0.134	0.120	0.140	0.137	0.128	0.271	0.178

Notes: Select results for Table 5 are reported for different model and parameter specifications. The units are billions of 2014 dollars or proportions. Results correspond to the 1970–2100 (exhaustion) timespan. Specifications are (1) the baseline specification; (2) baseline, but with the limit on the proportion of reserves extractable in a given year changed to $\max\{s_t, 2\%\}$; (3) baseline, but with a no limit on the proportion of reserves extractable in a given year; (4) baseline, but using a P90 reserve measure; (5) baseline, adding the distortionary tax items in Table A1 to costs; (6) has behavior computed with the competitive solution with wedge inclusive costs, but the costs of a particular allocation are evaluated with respect to economic costs only; (7) baseline, but restricting the sample to include only fields in active production in 1970; (8) baseline, but constraining fields to be usable in and after the first year of observed production, rather than discovery.

Table 7: Dynamic Counterfactual Results, Alternate Specifications.

Table 6: Static Counterfactual for 2014: Top 20 Producers.

Read. Table 6 shows that, in the 2014 static counterfactual, Persian Gulf OPEC producers would sharply expand output shares, especially Saudi Arabia and Kuwait. Table 7 shows that the dynamic results are robust across reserve, tax, extraction-limit, and sample restrictions.

Economic message. The efficient allocation is not a small perturbation of actual production. The counterfactual reallocates large production shares toward the Persian Gulf, and the main results survive many alternative assumptions.

11.8 Table 8: Dynamic Counterfactual Results, Conditional on Inferred Wedges and Figure 8: Costs over the Field Lifecycle

	Specification			
	(1)	(2)	(3)	(4)
Actual (A)	2,498	2,492	2,670	2,596
Counterfactual (C_1)	1,757	1,757	—	—
Counterfactual (C_2)	—	—	1,825	2,452
Total distortion ($A - C_1$)	741	735	—	—
Second-best distortion ($A - C_2$)	—	—	845	144
Distortion due to OPEC	—	—	845	144
Upper bound	174	158	—	—
Lower bound	117	100	—	—

Notes: The units are billions of 2014 dollars. Results correspond to the 1970–2100 (exhaustion) timespan. Specifications are: (1) the baseline specification; (2) baseline, but with no limit on the proportion of reserves extractable in a given year; (3) average over 20 iterations of a counterfactual computed conditional on inferred wedges for non-OPEC field-years, and wedges for OPEC reserves sampled (with replacement) i.i.d from all non-OPEC inferred wedges; (4) average over 20 iterations of a counterfactual computed conditional on inferred wedges for non-OPEC field-years, and wedges for OPEC reserves sampled (with replacement) i.i.d from all non-OPEC inferred wedges inferred for fields with c_f within \$5 of the OPEC field's c_f . The actual costs differ across specification due to different extraction rate assumptions in columns 1 and 2 and differences in wedges sampling between 3 and 4. This means that each specification has slightly different extraction paths post-2014. In keeping with the necessary structure of inferring wedges, all paths of extraction are computed taking costs as $c_f \mu_{it}$, but to minimize departures from the results in the rest of the paper, resource costs (in each specification) are evaluated using cost as measured by $c_f \mu_{it}$.

Table 8: Dynamic Counterfactual Results, Conditional on Inferred Wedges.

Read. Table 8 studies counterfactuals conditional on inferred wedges that rationalize non-OPEC deviations from perfect competition. Figure 8 shows cumulative costs relative to cumulative production over field life cycles.

Economic message. The inferred-wedge exercise addresses second-best concerns. Figure 8 helps show that start-up costs are present but do not overturn the main allocation logic.

11.9 Figure 9: Observed and Predicted Marginal Cost Ghawar Uthmaniyah (SA) and Figure 10: Deviation of Marginal Cost Specification and Output

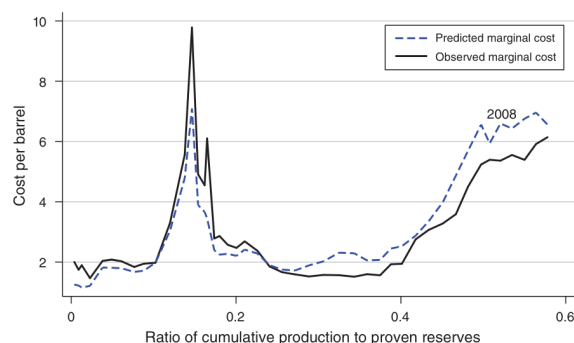


FIGURE 9. OBSERVED AND PREDICTED MARGINAL COST GHAWAR UTHMANIYAH (SA)

Observed and predicted marginal cost, using the cost specification in equation (13), is plotted against production. The vertical line indicates the proven reserves, and we insert the production year 2008, highest oil price in the sample period 1970–2014.

Figure 9: Observed and Predicted Marginal Cost Ghawar Uthmaniyah (SA).

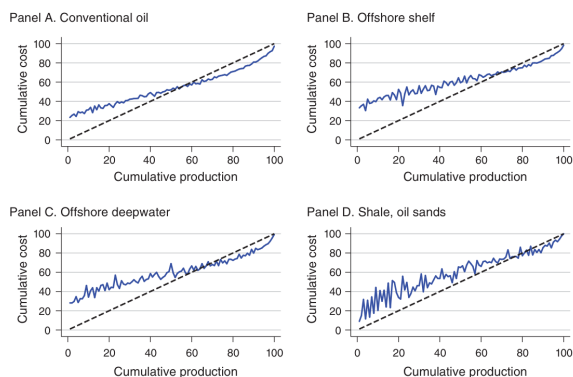


FIGURE 8. COSTS OVER THE FIELD LIFECYCLE

Notes: Cumulative production is measured as cumulative production for a field in year t divided by the total production observed over a field's lifespan. Cumulative costs are defined likewise for costs. Conventional oil is 72 percent of production, offshore shelf is 21 percent, offshore deepwater is 6 percent, and shale is 1 percent. Only fields that start producing after 1970 are used for this figure.

Figure 8: Costs over the Field Lifecycle.

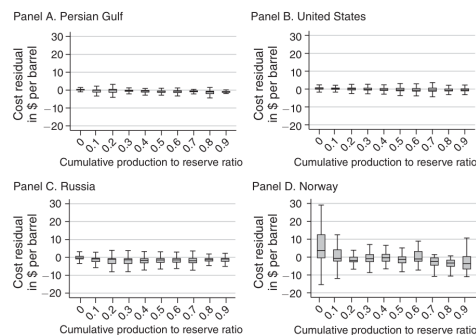


FIGURE 10. DEVIATION OF MARGINAL COST SPECIFICATION AND OUTPUT

Notes: The residuals from the cost specification in equation (13) are plotted against quantiles of the ratio of cumulative output-to-reserves, and weighted by the production of a field in total (country-level) production. The whiskers present the upper and lower adjacent values, while the box shows the twenty-fifth, fiftyth, and seventy-fifth percentile of the distribution respectively. We consider only fields with reserves that are equal to or higher than total recorded production over the observed life cycle of a field.

Figure 10: Deviation of Marginal Cost Specification and Output.

Read. Figure 9 compares observed and predicted marginal costs for the Ghawar Uthmaniyah field. Figure 10 plots residuals from the cost specification against cumulative production-to-reserve ratios. **Economic message.** These figures support the paper’s cost-function approximation. The constant marginal-cost assumption with technology-year shifts appears reasonable for the main counterfactual exercise.

11.10 Table A1: Definitions of Cost Components

TABLE A1—DEFINITIONS OF COST COMPONENTS	
Exploration capital expenditures	Costs incurred to find and prove hydrocarbons: seismic, wildcat, and appraisal wells, and general engineering costs.
Well capital expenditures	Capitalized costs related to well construction, including drilling costs, rig lease, well completion, well stimulation, steel costs, and materials.
Facility capital expenditures	Costs to develop, install, maintain, and modify surface installations and infrastructure.
Abandonment cost	Costs for decommissioning a field.
Production operating expenditures	Operational expenses directly related to the production activity. The category includes materials, tools, maintenance, equipment lease costs, and operation-related salaries. Depreciation and other non-cash items are not included.
Transportation operating expenditures	Represents the costs of bringing the oil and gas from the production site/processing plant to the pricing point (only upstream transportation). The category includes transport fees and blending costs.
SGA operating expenditures	Operating expenses not directly associated with field operations. The category includes administrative staff costs, office leases, related benefits (stocks and stock option plans), and professional expenses (legal, consulting, insurance). Only exploration and production-related SG&A are included.
Taxes operating expenditure	Local US taxes that are directly related to production. The category includes ad valorem taxes (county-based) and severance taxes (state-based).
Royalties	The sum of all gross taxes, including royalties and oil and export duties.
Government profit oil	The production-sharing agreement equivalent to petroleum taxes, but paid in kind (that is, the government contracts with a company to develop and operate the field, but retains rights to a proportion of the production). Government Profit Oil reduces the company’s entitlement production and is treated as a royalty effect in company reports.

Source: Rystad U-Cube External Use Documentation

Table A1: Definitions of Cost Components.

Read. Table A1 defines the cost categories used in the Rystad data, including exploration capital expenditures, well capital expenditures, facility capital expenditures, abandonment costs, production operating expenditures, transportation operating expenditures, SG&A, taxes, royalties, and government profit oil.

Economic message. Clear cost definitions are central to the paper’s contribution. The ability to map observed field-level expenditures into resource costs is what makes the cost-based misallocation approach operational.

12 Conclusion

Asker, Collard-Wexler, and De Loecker (2019) show that misallocation can be measured directly through production costs when detailed field-level cost data exist. In the global oil market, low-cost OPEC fields, especially Saudi and Kuwaiti fields, produce less than they would in a competitive allocation, while higher-cost fields elsewhere produce more. This creates a large welfare-relevant resource-cost distortion. The paper’s broader message is that market power can matter through the allocation of production across heterogeneous units, not only through reductions in total output.

For future research, the paper is a strong template for measuring how institutions, market power, taxes, and political frictions shape real resource allocation.